

Madelon Hulsebos

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I'm a computer scientist and AI researcher working on the democratization of insights from tabular data (e.g. relational tables and spreadsheets) through tabular representation learning (TRL). My research involves machine learning, natural language processing, information retrieval and data management.

EMPLOYMENT

Centrum Wiskunde & Informatica (CWI) – Amsterdam, Netherlands	
<i>Tenured Research Faculty, PI, lead of the TRL Lab</i>	2026–present
<i>Tenure-Track Research Faculty, PI, lead of the TRL Lab</i>	2024–2026
University of California, Berkeley – Berkeley, United States	
<i>Postdoctoral Scholar, EECS and BIDS</i>	2023–2024
University of Amsterdam – Amsterdam, Netherlands	
<i>PhD Researcher, Informatics Institute</i>	2023
<i>Guest Researcher, Informatics Institute</i>	2020–2023
Sigma Computing – San Francisco, United States	
<i>PhD Researcher (full-time)</i>	2021–2022
<i>PhD Research Intern</i>	2021
KPN/Heineken – Rotterdam/Amsterdam, Netherlands	
<i>Data Scientist</i>	2019–2021
Massachusetts Institute of Technology – Cambridge, United States	
<i>Research Visitor, MIT Media Lab</i>	2018–2019
Delft University of Technology – Delft, Netherlands	
<i>Graduate TA, Pattern Recognition & Web Information Systems groups</i>	2017–2018
Aalto University – Helsinki, Finland	
<i>Research and Teaching Assistant, Machine Learning for Big Data</i>	2017

EDUCATION

University of Amsterdam – Amsterdam, Netherlands	
<i>PhD Computer Science</i>	2020–2024
Delft University of Technology – Delft, Netherlands	
<i>MSc Computer Science</i>	2016–2018
<i>BSc Technology, Policy and Management</i>	2011–2015

ANCILLARY ACTIVITIES

ELLIS unit Amsterdam – Amsterdam, Netherlands	
<i>Member of the Scientific Management Team</i>	2026–present
Prior Labs – Berlin/Freiburg, Germany	
<i>Member of the Scientific Advisory Board</i>	2025–present
UniPartners Delft – Delft, Netherlands	
<i>Member of the Supervisory Board</i>	2017–2023
<i>Member of the Executive Board</i>	2015–2016

ACADEMIC SERVICE

Chairing

Founder and co-chair, Table Representation Learning workshop (TRL) @ NeurIPS, ACL	2022–2025
Co-chair, AI for Tabular Data workshop @ EurIPS	2025
Co-chair Tutorial Track @ VLDB	2025

Co-chair, New Researcher Symposium @ SIGMOD	2025
Co-chair Seminar on Challenges and Opportunities in Table Representation Learning @ Dagstuhl	2025
Co-chair Representation Learning and Generative Models for Structured Data @ ELLIS Amsterdam	2025
Initiator, Table Representation Learning research theme @ ELLIS Amsterdam	2025
Co-chair, Data Management for End-to-End ML workshop (DEEM) @ SIGMOD	2023-2025
Steering Committee, Tabular Data Analysis workshop (TaDA) @ VLDB	2023, 2024
Co-chair, SemTab challenge @ ISWC	2021-2023

Conference/Journal Reviewing

EMNLP Industry Track	2026
KDD	2026
ACL ARR	2026
NeurIPS	2026
Conference of Innovative Data Systems Research (CIDR)	2025-2026
SIGMOD	2026
VLDB Journal	2025-2026
NeurIPS D&E track	2022-2025
PVLDB	2024-2026
ICDE (Industry track)	2024
EDBT (Industry track)	2022-2023
TheWebConf (Industry track)	2022-2023

Workshop Reviewing

BioDMS@VLDB'26, NOVAS @ SIGMOD'25, aiDM @ SIGMOD'24, DMLR @ICML'23-'24, DBML @ ICDE'23-'24, PhD workshop @ VLDB'23, SemTab @ ISWC'21-'23, AIDB @ VLDB'22.

Editorial

Assistant Editor, Journal of Systems Research (JSys)	2022 - 2023
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ADVISING AND COMMITTEES

Advising postdoc/PhD

Félix Lefebvre, CWI, postdoc	2026-
Dennis Loevlie, CWI, PhD	2026-
Rohith Prabakaran, CWI, PhD	2026-
Aécio Santos, CWI, Postdoc	2025-
Xue (Effy) Li, CWI, Postdoc	2025-
Cornelius Wolff, CWI, PhD	2025-
Daniel Gomm, CWI, PhD	2024-

PhD Interns

Nour Shaheen, Mila / Polytechnique Montreal	2026
Giulia Perciballi, University of Sorbonne	2026
Allaa Boutaleb, LIP6, University of Sorbonne	2026

Advising (under)graduate

Hongqian Xia, University of Amsterdam, MSc (now at Booking)	2026
Lisa van Oosten, University of Amsterdam, MSc (unknown)	2026
Yme Kingma, Radboud University, MSc (unknown)	2026
Louis Gehringer, University of Amsterdam, MSc (now at startup)	2026
Ahmed Omar Eissa, Utrecht University, MSc (now at Uber)	2025-2026
Daniel De Dios Allegue, TU Delft, pre-MSc (now at Cambridge)	2025-2026
Wojciech Kosiuk, University of Amsterdam, MSc (now at Mistral)	2025-2026
Jan Henrik Bertrand, University of Amsterdam, MSc (now at startup)	2025-2026
Liang Telkamp, University of Amsterdam, MSc (🏆 Amsterdam AI Thesis Award) (now at UN)	2025
Nikola Selic, TU Munich, MSc (unknown)	2025
Wenjing Lin, UC Berkeley, MSc (now at WuXi AppTec)	2024-2026
Rachel Lin, UC Berkeley, MSc (now at Assort)	2024-2025
Xingyu Ji, UC Berkeley, BSc (now at Nooks)	2023-2025
Teun Mathijssen, University of Amsterdam, MSc (unknown)	2023

Thesis Committees

Irina Saparina, University of Edinburgh, PhD	2026
Lukas Lakowski, HPI, PhD	2026
Tanguy Urvoy, University of Rennes, HDR	2026
Lennart Behme, BIFOLD/TU Berlin, PhD	2026
Félix Lefebvre, Inria, PhD	2026
Margherita Martorana, VU Amsterdam, PhD	2025
Vaishali Pal, University of Amsterdam, PhD	2025
Tianji Cong, University of Michigan, PhD	2023-2025

AWARDS AND FUNDING (€2.3M+)

Compute/model credit grant, \$35K, Google	2026
Table Representation Learning Theme at ELLIS unit Amsterdam, €50K, SAP	2026
Research grant for collaboration on TRL CWI x Inria, €727K, CWI	2025
Model credit grants, \$7K, OpenAI & Cohere	2025
Amsterdam AI Thesis Award for the supervised MSc thesis by Liang Telkamp	2025
Table Representation Learning Theme at ELLIS unit Amsterdam, €50K, SAP	2024
Table Representation Learning Research Grant, €300K, SAP	2024
AiNed Fellowship Grant, €993K, NWO	2024
ELLIS Membership	2024
Postdoctoral Fellowship, \$150K, Accenture-BIDS	2023
Best Reviewer Award, PhD Workshop, VLDB	2023
Travel Award, \$2.5K, VLDB Endowment	2022
Honorable mention GitTables, SemTab challenge	2021

TALKS, PANELS AND LECTURES**Invited Talks and Keynotes**

<i>On the Different Flavors of Tabular Reasoning</i>	
BIFOLD, TU Berlin	2026
Institute for Language, Cognition and Computation Seminar, University of Edinburgh	2026
AI Research Retreat, SAP	2026
<i>What AI is Leaving on the Table</i>	
Deep Learning Extravaganza, ELLIS Amsterdam	2026
AI for Good Webinar “Discovery: From Molecules to Models”, ELLIS & UN ITU	2026
<i>Towards Contextual Sensitive Data Detection</i>	
ICMD Consultation, Raoul Wallenberg Institute, Sweden	2026
UNECE Expert Meeting on Statistical Data Confidentiality, Spain	2025
<i>What are we asking from Tabular Data? [keynote]</i>	
AI for Tabular Data workshop, EurIPS, Denmark	2025
<i>Towards robust open-domain querying over structured data</i>	
KIK AI seminar, Amsterdam University Medical Center, Netherlands	2025
ELLIS NL Scientific Day, Netherlands	2025
AI Research Retreat, SAP, Germany	2025
Business Analytics Seminar, University of Amsterdam, Netherlands	2025
Seminar on AI, Semantics, and Tabular Data, IBM, USA	2025
<i>What Table Representation Learning brings to Data Systems</i>	
COMPASS Seminar, ETH, Zürich, Switzerland	2024
CWI Lectures, CWI, Amsterdam, Netherlands	2024
<i>Retrieval Systems for Structured Data: the missing piece for grounding LLM-powered query interfaces in factual data</i>	
Berkeley Institute for Data Science Seminar, Berkeley, USA	2024
Berkeley EECS EPIC Retreat, Berkeley, USA	2024
Harnessing AI for Relational Data: Industry and Research Perspectives, SF, USA	2024
<i>Advances, challenges, and opportunities in Table Representation Learning</i>	
Microsoft Gray Systems Lab, USA	2024
University of Washington Seattle, Seattle, USA	2024

Snowflake, USA	2024
Google, Sunnyvale, USA	2024
UC Berkeley, Berkeley, USA	2024
Transformers at Work 2023, Zeta Alpha, Amsterdam, Netherlands	2023
<i>Towards Table Representation Learning for end-to-end data management and analysis</i>	
INRIA-Saclay, Paris, France	2023
ML for Systems and Systems for ML Workshop @ BTW, Dresden, Germany	2023
Hasso Plattner Institute, Berlin, Germany	2023
TU Darmstadt, Darmstadt, Germany	2022
<i>GitTables: a large corpus of relational tables</i>	
Tabular Data Analysis workshop, VLDB, Vancouver, Canada	2023
Database Architectures group, CWI, Amsterdam, Netherlands	2022
(Guest) Lectures	
<i>Representation Learning and Generative Models for Tabular Data</i>	
University of Amsterdam	2026
BIFOLD, TU Berlin	2025
Panel discussions	
“Neural Relational Data: Tabular Foundation Models, LLMs... or both?”, VLDB, London	2025
Transformers at Work 2023, Zeta Alpha, Amsterdam, Netherlands	2023

PUBLICATIONS

2026

The Silent Failures in Agentic Data Science, **Under Review**

Li, X., Xia, H., Gomm, D., Santos, A., Wolff, C., [Hulsebos, M.](#)

Tabular Selective Attention, **Under Review**

Prabakaran, R., Allegue, D., Shaheen, N., Holzmüller, D., [Hulsebos, M.](#)

SQaLe: A Large Realistic Dataset to Empower Small Specialized Text-to-SQL, **Under Review**

Wolff, C., Gomm, D., [Hulsebos, M.](#)

Open Tabular Insight Extraction: Where do we Stand and Where do we Go?, **Under Review**

Gomm, D., De Rijke, M., Hulsebos, M.

Fine-Grained Table Retrieval Through the Lens of Complex Queries, **AKBC at EMNLP**

Kosiuk, W., Ji, X., Chung, Y., Özcan, F., [Hulsebos, M.](#)

Towards Contextual Sensitive Data Detection, **Findings of EMNLP**

Telkamp, L., [Hulsebos, M.](#)

2025

SQaLe: A Large Text-to-SQL Corpus Grounded in Real Schemas, **AI for Tabular Data workshop at EurIPS 2025**

Wolff, C., Gomm, D., [Hulsebos, M.](#)

Are We Asking the Right Questions? On Ambiguity in Natural Language Queries for Tabular Data Analysis, **AI for Tabular Data workshop at EurIPS 2025**

Gomm, D., Wolff, C., [Hulsebos, M.](#)

Unlocking the Full Potential of Data Science Requires Tabular Foundation Models, Agents, and Humans

Cong, T., Eisenschlos, J., Gomm, D., Grinsztajn, L., Müller, A., Sanghi, A., et al., Binnig, C., [Hulsebos, M.](#), Hutter, F.

How well do LLMs reason over tabular data, really?, **Table Representation Learning workshop at ACL**

Wolff, D., [Hulsebos, M.](#)

Rethinking Dataset Discovery with DataScout, **User Interface Software and Technology (UIST)**

Lin, R., Chopra, B., Lin, W. Shankar, S. [Hulsebos, M.](#), Parameswaran, A.

Metadata Matters in Dense Table Retrieval, **ELLIS workshop on RL and GM for Structured Data**
Gomm, D., [Hulsebos, M.](#)

TARGET: Benchmarking Table Retrieval for Generative Tasks, **In Review**
Li, X., Glenn, P., Parameswaran, A., [Hulsebos, M.](#)

Towards Accurate and Efficient Document Analytics with Large Language Models, **ICDE**
Lin, Y., [Hulsebos, M.](#), Ma, R., Shankar, S., Zeigham, S., Parameswaran, A. G., & Wu, E.

2024

TARGET: Benchmarking Table Retrieval for Generative Tasks, **TRL @ NeurIPS**
Li, X., Parameswaran, A., [Hulsebos, M.](#)

"It Took Longer than I was Expecting:" Why is Dataset Search Still so Hard?, **HILDA@SIGMOD**
[Hulsebos, M.](#), Lin, W., Shankar, S., Parameswaran, A.

SchemaPile: A Large Collection of Relational Database Schemas, **SIGMOD**
Doehmen, T., Geacu, R., [Hulsebos, M.](#), Schelter, S.

SPADE: Synthesizing Assertions for Large Language Model Pipelines, **Proceedings of VLDB**
Shankar, S., Li, H., Asawa, P., [Hulsebos, M.](#), Lin, Y., Zamfirescu-Pereira, J., Chase, H., Fu-Hinthorn, W.,
Parameswaran, A., Wu, E.

Eighth Workshop on Data Management for End-to-End Machine Learning (DEEM), **SIGMOD**
[Hulsebos, M.](#), Shankar, S., Interlandi, M.

2023

AdaTyper: Adaptive Semantic Type Detection, **ArXived**
[Hulsebos, M.](#), Groth, P., Demiralp, C.

Introducing the Observatory Library for End-to-End Table Embedding Inference, **TRL @ NeurIPS**
Cong, T., Sun., Z., Groth, P., Jagadish, H., [Hulsebos, M.](#)

Observatory: Characterizing Embeddings of Relational Tables, **Proceedings of VLDB**
Cong, T., [Hulsebos, M.](#), Sun., Z. Groth, P., Jagadish, H.V.

Models and Practice of Neural Table Representations [tutorial], **SIGMOD**
[Hulsebos, M.](#), Deng, X., Sun, H., Papotti, P.

Seventh Workshop on Data Management for End-to-End Machine Learning (DEEM), **SIGMOD**
Boehm, M., [Hulsebos, M.](#), Shankar, S., Varma, P.

2022

GitTables: A Large-Scale Corpus of Relational Tables, **SIGMOD**
[Hulsebos, M.](#), Demiralp, C., Groth, P.

GitSchemas: A Dataset for Automating Relational Data Preparation Tasks, **DBML @ ICDE**
Döhmen, T., [Hulsebos, M.](#), Beecks, C., Schelter, S.

Results of SemTab 2022, **Proceedings of SemTab @ ISWC**
Abdelmageed, N., Chen, J., Cutrona, V., Efthymiou, V., Hassanzadeh, O., [Hulsebos, M.](#), Jiménez-Ruiz, E.,
Sequeda, J. and Srinivas, K.

Making Table Understanding Work in Practice [abstract], **CIDR**
[Hulsebos, M.](#), Gathani, S., Gale, J., Dillig, I., Groth, P., Demiralp, C.

Augmenting Decision Making via Interactive What-If Analysis, **CIDR**
Gathani, S., [Hulsebos, M.](#), Gale, J., Haas, P. J., Demiralp, C.

2021

Results of SemTab 2021, Proceedings of SemTab @ ISWC

Cutrona, V., Chen, J., Efthymiou, V., Hassanzadeh, O., Jiménez-Ruiz, E., Sequeda, J., Srinivas, K., Abdelmageed, N., Hulsebos, M., Oliveira, D., Pesquita, C.

2020

Sato: Contextual semantic type detection in tables, Proceedings of VLDB

Zhang, D., Suhara, Y., Li, J., Hulsebos, M., Demiralp, C., Tan, W.

2019

Sherlock: A deep learning approach to semantic data type detection, ACM SIGKDD

Hulsebos, M., Hu, K., Bakker, M., Zraggen, E., Satyanarayan, A., Kraska, T., Demiralp, C., Hidalgo, C.

VizNet: Towards a large-scale visualization learning and benchmarking repository, ACM CHI

Hu, K., Gaikwad, N., Hulsebos, M., Bakker, M., Zraggen, E., Hidalgo, C., Kraska, T., Li, G., Satyanarayan, A., Demiralp, C.

2018

The Network Nullspace Property for Compressed Sensing of Big Data Over Networks, IEEE ICASSP

Hulsebos, M., Jung, A.